

CAPSTONE – II

PROGRESS REPORT

Project Name: Sentiment Analysis using Hierarchical Reasoning Models and Mixture-of-Experts

Architecture: Experimental Results and Analysis

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Tasks Completed:

1. Deep Learning Pipeline:

- *Extricated the core inference `Pipeline` inside `Code/models/models.py` resulting in a strictly mathematical representation to calculate the feasibility of advance architectures like transformers, hrm's, mixture-of-experts architecture.*
- *Refactored dynamic configurations and path resolvers natively into module initialization rather than system constants to increase isolation.*

2. Training Refactoring (TRAIN-TESTS):

- *Overhauled deep learning scripts across the `Code/train` directories (`train.py`, `finetune\_pipeline.py`, `assemble\_pipeline.py`), transitioning them to operate directly upon the localized, decoupled `Pipeline`.*
- *Re-calibrated the full regression validation, hypothesis, gradient integration, and edge-case test suites under `Code/test/`. Deleted obsolete state tests and corrected scoped testing frameworks to achieve an error-free (Exit Code 0) validation baseline against the isolated model operations.*

3. HRMs and Mixture-of-Experts (MOEs):

- *Refactored internal `HRMPipelineBuilder` components to inherently spawn decoupled pipelines out of the box.*
- *Removed obscure runtime kwargs injection dependencies, ensuring Hierarchical Reasoning Models construct exclusively off verifiable Mermaid definitions with transparent network paths suitable for MoE layers.*

4. Data Preprocessing Enhancements:

- *Validated batch dimensionality resilience comprehensively by introducing property-based hypothesis testing workflows on dynamic inputs constraints, verifying stable data-loader scaling across variable batch sizes.*

Total Hours: 37 hours (1-weeks)

Plan for Next Week:

1. Distributed Model Training Implementations:

- *Execute scalable multi-GPU implementations relying on the pristine pipeline states across the decoupled MoE-HRM system environments.*

2. Empirical Evaluations and Sentiment Inference:

- *Standardize inferencing APIs directly on sentiment extraction tasks to generate quantifiable scoring analytics on validation data subsets.*